

TimeTensors Experiment 2: Sampling and Batch Size

1 Theoretical overview and goal

Time-series windows are correlated along both user and date axes. A mini-batch therefore defines more than a computational unit: it determines which sources of variation contribute to a gradient. Random-window sampling approximates the global empirical risk, date sampling emphasizes cross-sectional diversity, and individual sampling emphasizes temporal diversity within selected users.

Experimental goal. Measure how user/date diversity and gradient batch size affect convergence and final generalization under a fixed optimization budget.

2 Implementation details

The modes are `random`, `dates`, `individuals`, and `all`. The exhaustive profile crosses them with batch sizes 64, 256, and 1024. The exhaustive mode deterministically exposes every accessible window; other modes draw according to their named axis. Evaluation sampling remains fixed so only training exposure changes.

The default study holds six datasets, three (L, H) settings, DLinear, seeds 1–3, learning rate 10^{-5} , 10,000 optimizer steps, and 1,000-step validation constant. It compares random sampling at batch sizes 64, 256, and 1024 with individual sampling at batch size 256. The full profile crosses all four modes with all three batch sizes. The launcher is `02_sampling.slurm`.

Report final MSE with seed standard deviation, but also retain interval-average train loss and validation curves. Because different sampling modes expose different numbers of unique windows per optimizer step, include the effective dataset size and number of optimizer updates when interpreting equal-epoch runs.

3 Interpretation

A batch-size effect can be confounded with diversity: a large batch from one user is not equivalent to a smaller cross-user batch. Conclusions should therefore compare batch size within each mode first, then compare the best stable configuration across modes.